

# Chapter 16

## Basics of Research Report Writing for Behavioural Science Students and Emerging Scholars

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### ABSTRACT

*This chapter is written with behavioural science students and emerging scholars in mind. Behavioural science in this context includes but is not limited to management sciences, social sciences, psychology, education, and any discipline that deals with human actions. The chapter is written to sharpen the knowledge and skills of students in these areas in writing a good quality research report (at advanced undergraduate level), thesis, or dissertation (at postgraduate level). The chapter has valuable hints that will guide behavioural science students and emerging scholars in writing good quality research reports, theses, or dissertations. The hints will also be helpful for supervisors and mentors in guiding and assessing their student's work. A research project, thesis, or dissertation is essential for graduation from tertiary institutions. The need, therefore, to write good quality research reports requires that students possess the necessary knowledge and skills in writing.*

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*Basics of Research Report Writing for Behavioural Science Students and Emerging Scholars***CHAPTER SUMMARY**

This chapter was written primarily to guide emerging scholars, students, and early career researchers on the basic constituents of a research report for a thesis, projects, dissertation and other related documents. The chapter was organised to capture the basic elements in the preliminary pages of a research report, the sub-components of chapters 1, 2, 3, 4 and 5, as well as supplementary pages. The preliminary pages comprised the title page, declaration, certification, acknowledgement, abstract, tables of contents, list of tables and list of figures. Chapter one provides the basic idea or problem underlying the research, including a justification of its needs based on existing studies in the literature. Crucial research questions and/or hypotheses (for quantitative studies) underlying the study are posed in this chapter. In chapter two, an extensive and detailed review of relevant and related literature is reviewed. It is also in chapter two that all relevant gaps in methodology, evidence, theory, knowledge, and population are identified to enable the researcher to find a base for his/her study. In chapter three, the methods, materials, and procedures are presented to achieve the goals (broad and specific) earlier stated in chapter 1. Chapter four is concerned with the presentation and interpretation of results as well as the discussion of key findings. Chapter five presents the summary of the entire research process from chapters one to four. Conclusions reached based on the results in chapter four are mentioned in chapter five, including the study's theoretical, practical and research implications. Finally, suggestions for further research are made in chapter five based on the study's limitations.

**INTRODUCTION**

The theme of this chapter is twofold; hints that will guide behavioural science students in writing good-quality research projects, theses and dissertations. The hints will also be helpful for supervisors and mentors in guiding and assessing their students' work. Behavioural science in this paper includes but is not limited to management sciences (e.g., management, marketing, etc.), social sciences, education and any discipline that deals with human actions. The need to adopt best practices in research reports and supervise students' theses motivated the writing of this chapter.

Problems exist in every sphere of life, and knowledge is incomplete in every discipline. Many questions require answers to solve numerous problems in the world. Gaps in our knowledge and unresolved problems can be addressed by asking relevant questions and seeking answers through research (Swales & Feak, 2014). Research is a fundamental tool for decision-making in most fields of endeavours (Kerlinger, 2000). Therefore, researchers, including tertiary-level students, emerging scholars,

***Basics of Research Report Writing for Behavioural Science Students and Emerging Scholars***

supervisors, and academic mentors, must understand what to write and how to write and research the chosen topic in their fields for graduation.

The researcher who completes his/her research must present a written report using a good style and format. In writing a research report, the objectives should be evident in the researcher's mind. S/He should understand that his/her presentation should achieve these specific objectives (Isangedighi & Bassey, 2012).

1. Research reports are written to acquaint the reader with the research problem. Through reading the report, the reader should be clear about what the researcher had in mind regarding the problem, his conception, and what the problem implied. Such clarity will help the reader to orient himself properly while reading the work;
2. Reports are written to fully and comprehensively outline the research data. Such data should be presented to see the basis for results interpretation and conclusions. The data should be so precisely presented that anyone who wants to replicate the study is adequately armed. The data should be logically presented. This is seen in the precise description of the data gathered, their coding and utilisation in addressing each subset of the problem;
3. reports are written to show appropriate data interpretation and the implications of the data in solving the problem. This means that giving research data is not enough. The mind must be allowed to battle with the available data to bring out meaning. It is only when the mind inquiries into the innate meaning of the data so assembled that the problem is said to have been resolved.

It has, however, been observed that researchers, students and emerging scholars need to adopt better practices in research report writing. This agrees with the position of Ugodulunwa (2016) that there are common pitfalls in many thesis proposals for defence and manuscripts submitted for review in journals. These include a preliminary literature review; poorly written abstract; unclear introduction; ambiguous research questions and hypotheses; inadequate description of sample and sampling procedure; inappropriate methodology/use of wrong statistical techniques; inappropriate description of instruments for data collection; poor presentation of results, discussion and conclusion. They also indulge in over-generalisation of findings, and poor writing and citation styles, among others. This chapter is, therefore, motivated by the need to sharpen students' knowledge and skills in writing acceptable research reports for undergraduate projects, theses, and dissertations and to provide guidance for supervising students' research reports for graduation. This chapter is written to satisfy that need.

At the end of this chapter, readers should be able to:

***Basics of Research Report Writing for Behavioural Science Students and Emerging Scholars***

1. define the term 'research';
2. explain the usefulness of research guidelines in the writing and supervision of scholarly works;
3. outline the basic features of a suitable research project format;
4. discuss the major areas in the research presentation format outlined in (iii) above;
5. write a good research proposal, project, and thesis following the format in this chapter.

**FORMAT FOR UNDERGRADUATE PROJECT, THESIS OR DISSERTATION**

This section focuses on the format for organising and presenting reports of students' research projects, theses and dissertations in management sciences, social sciences education, and humanities. Generally, the format of the reports serves as a guide to researchers/students. Different institutions and faculties adopt various formats, and any of them serves the same purpose as long as the fundamentals of research report details are included. Therefore, students and emerging scholars are advised to adhere strictly to the required format of their institution, funding agency or publication outlet as much as possible.

Research projects, theses or dissertation reports writing are usually structured or formatted into three parts, namely:

- The Preliminary Pages (usually numbered in Roman Numerals)
- The Main Body (consisting of five chapters)
- The Ending Pages (usually numbered in Arabic Numerals)

The components of each section, including what the researcher or student is expected to say under each component, are given below:

**PRELIMINARY PAGES (USUALLY NUMBERED IN ROMAN NUMERALS)****i. Cover Page**

This is usually a softbound cover for the proposal and thesis before the defence. It is often transformed into a hardcover page ready for submission after a successful defence. The colour of the cover is determined by the institution, type of degree

***Basics of Research Report Writing for Behavioural Science Students and Emerging Scholars***

or student's department. It must contain the title of the thesis, the student's name, registration numbers, the department and the month and year of successful defence, depending on the institution's style.

**ii. Title Page**

The title page contains information on the project title, thesis or dissertation, students' name, degree in view, name of the institution, and the date (month and year) of submission of the report. An example of a cover page is shown in Figure 1; however, this might vary with institutions.

*Figure 1.*

**CONFLICT MANAGEMENT STRATEGIES AND SECONDARY SCHOOL  
TEACHERS' JOB EFFECTIVENESS IN OBUBRA LOCAL  
GOVERNMENT AREA OF CROSS RIVER STATE,  
NIGERIA.**

**A RESEARCH PROJECT**

**BY**

**OWAN, VALENTINE JOSEPH  
MATRIC. NO. 15/063143057**

**SUBMITTED TO**

**DEPARTMENT OF EDUCATIONAL ADMINISTRATION  
AND PLANNING, FACULTY OF EDUCATION  
UNIVERSITY OF CALABAR  
CALABAR, NIGERIA.**

**IN PARTIAL FULFILMENT OF THE REQUIREMENTS FOR THE AWARD  
OF BACHELOR OF EDUCATION DEGREE (B.Ed.) IN EDUCATIONAL  
ADMINISTRATION AND PLANNING.**

**SEPTEMBER, 2018.**

**iii. Declaration Page (by the Student)**

Here, the student declares or attests that the research is an original work; appends his/her signatures with the date. The implication is that if it is discovered that there

**Basics of Research Report Writing for Behavioural Science Students and Emerging Scholars**

was plagiarism over time, the student becomes liable for committing academic fraud. An example of a declaration page is shown in Figure 2.

Figure 2.

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**DECLARATION**

I Owan, Valentine Joseph an undergraduate student of the Department of Educational Administration and Planning, Faculty of Education, University of Calabar, Calabar, with Matriculation Number 15/063143057, declare that this research work on “Conflict Management Strategies and Secondary School Teachers’ Job Effectiveness” is an original study carried out in Obubra Local Government Area of Cross River State in 2017/2018 academic session. I hereby also declare that the work has not been duplicated in parts or in full for any other Diploma or Degree programme.

|                                                                                                                                                                                                                                                           |                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <hr style="border: 0; border-top: 1px solid black; margin-bottom: 5px;"/> <p><b>Owan, Valentine Joseph</b><br/>(Student)<br/>Department of Educational<br/>Administration and Planning,<br/>Faculty of Education,<br/>University of Calabar, Calabar.</p> | <hr style="border: 0; border-top: 1px solid black; margin-bottom: 5px;"/> <p style="text-align: center;">Date</p> |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|

#### iv. Certification Page (by the Examiners)

Here, all the people that have the final say on the project, theses or dissertation certify by appending their signatures and date. This includes the supervisor (s), head of department, graduate school representative, and external examiner. Their signatures and dates give credence that having read and examined the work has met the requirements for awarding a degree in a specialised field of endeavour. An example of a certification page is presented in Figure 3.

**Basics of Research Report Writing for Behavioural Science Students and Emerging Scholars***Figure 3.***CERTIFICATION**

This is to certify that this study on Conflict Management Strategies and Secondary School Teachers' Job Effectiveness in Obubra Local Government Area of Cross River State, is an original research work conducted by Owan, Valentine Joseph with Matriculation number 15/063143057 under my supervision. I hereby certify that this work has been approved as having met the requirements for the award of Bachelor of Education degree in the Department of Educational Administration and Planning, Faculty of Education, University of Calabar, Calabar.

**Prof. Charles P. Akpan**  
(Supervisor)  
Department of Educational  
Administration and Planning,  
University of Calabar, Calabar.

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Signature

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Date**v. Dedication PAGE (if Any)**

A research project, thesis or dissertation report, just like a textbook, may be dedicated to a particular person or group of persons - a spouse, children, parents or other family members, a group of students, etc. The choice depends on the author. It is, however, not a requirement.

**vi. Acknowledgments**

In this section, the researcher formally expresses appreciation to various individuals or groups who have contributed in one way or the other to the successful completion of the study. Such contributions may have been intellectual, financial, moral or spiritual. Since the research is an academic exercise, acknowledgement or persons in the academic circle should be acknowledged first before considering others. The order of acknowledgement could be God first, supervisor(s), head of department,

**Basics of Research Report Writing for Behavioural Science Students and Emerging Scholars**

lecturers and other scholars, family members, etc. A sample of acknowledgement is presented in Figure 4.

*Figure 4.*

#### **ACKNOWLEDGEMENTS**

I am most grateful to the Lord Almighty for granting me life and good health throughout my academic duration, and for the uncommon strength he gave to successfully complete this research work. I am particularly grateful to my supervisor Prof. Charles P. Akpan for his diligence, guidance, and constructive corrections he offered to ensure that the product of the study was standardized. May God grant you more strength and wisdom.

Specifically, it will be unwise if I fail to acknowledge the efforts of Assoc. Prof. Barr. Arop, Festus O; Assoc. Prof. B. A. Bassey; Dr. (Mrs). Odigwe, Francisca N.; Mr. Offem O. Otu; Mrs. Ene, Egbula Ogar; Assoc. Prof. James E. Okon; Prof. (Mrs). Uchendu C. C.; Dr. (Mrs.) R. O. Osim; Prof. M. O. Nyong; Dr. Ebi Bassey; Dr. Peter Ubi; Assoc. Prof. Ovat, O. O.; Mr. Ekaette Samuel O.; Mrs. Nwannunu Blessing Iheoma; and Mr. Wonah Fidelis, all of whom have been like parental figures to me throughout my time in the university, and for imparting positive intellectual and moral skills to my person.

My heartfelt gratitude, utmost joy, and intense honour goes to my parents Mr. and Mrs. Owan, Joshua N.; and my siblings Tina, David, Augustine, and Ruth Owan for their doggedness, labour, hard work, financial and moral support, prayers, encouragement and for inconveniencing themselves just to make me relevant in life. I am sincerely grateful to the entire family of Rev. Dr. Steve Owan for their whole-hearted love, care, prayer and support throughout the duration of my study.

## **vii. Abstract**

An abstract is a summary or a synopsis of the entire work, giving the reader a clear and precise idea of what was done. An abstract should briefly state the study's



***Basics of Research Report Writing for Behavioural Science Students and Emerging Scholars***

problems, the purpose of the study, the method of investigation, the significant findings are, as well as the major recommendations. An informative abstract of a research project, thesis or dissertation should be between 300-500 words or one and a half pages at most (Asim et al., 2017; Ndiyo, 2005). A sample of an abstract is presented in Figure 5.

*Figure 5.*

**ABSTRACT**

The study investigated conflict management strategies and secondary school teachers' job effectiveness in Obubra Local Government Area of Cross River State. Six null hypotheses were formulated to guide the study. The study adopted correlational and factorial research designs. Purposive sampling technique was used to select a sample of 222 teachers from a population of 352 secondary school teachers. Conflict Management Strategies Questionnaire (CMSQ) and Secondary School Teachers' Job Effectiveness Questionnaire (SSTJEQ) were used respectively, as instruments for data collection. The hypotheses were tested at .05 level of significance using Population t-test, Pearson Product Moment Correlation and Multiple Regression analyses. Findings revealed that, teachers' job effectiveness level in Obubra Local Government Area is significantly high. Findings also revealed that arbitration, dialogue, and effective communication strategies respectively, had a significant relationship with secondary school teachers' job effectiveness. Smoothing strategy had no significant relationship to secondary school teachers' job effectiveness. The findings also revealed among others that; the four conflict management strategies (arbitration, dialogue, effective communication and smoothing) had a joint significant influence on secondary school teachers' job effectiveness. Based on these findings, it was recommended among others that; secondary school principals should not rely totally on one conflict management strategy as the best for all situations, instead they should learn how to use various conflict management strategies, and apply them in any given conflict situation in their schools.

**Basics of Research Report Writing for Behavioural Science Students and Emerging Scholars****viii. Table of Contents**

This contains the title/sub-titles of every section/sub-section of the report, indicating the specific pages where each section and subsection can be located in the body of the report. It consists of a list of preliminary information, chapters and titles, as well as a list of tables, list of figures, a list of abbreviations, references/bibliography and an appendix. A sample of a table of contents is displayed in Figure 6.

*Figure 6.*

| <b>TABLE OF CONTENTS</b>         |     |
|----------------------------------|-----|
| TITLE PAGE                       | i   |
| CERTIFICATION                    | ii  |
| DECLARATION                      | iii |
| ACKNOWLEDGMENTS                  | iv  |
| ABSTRACT                         | vi  |
| TABLE OF CONTENTS                | vii |
| LIST OF TABLES                   | x   |
| <b>CHAPTER ONE: INTRODUCTION</b> |     |
| 1.1. Background to the study     | 1   |
| 1.2. Theoretical framework       | 10  |
| 1.3. Statement of the problem    | 13  |
| 1.4. Purpose of the study        | 15  |
| 1.5. Research questions          | 16  |
| 1.6. Statement of hypotheses     | 16  |
| 1.7. Significance of the study   | 17  |
| 1.8. Assumptions of the study    | 19  |
| 1.9. Scope of the study          | 19  |
| 1.10. Limitations of the study   | 20  |
| 1.11. Definition of terms        | 20  |

**Basics of Research Report Writing for Behavioural Science Students and Emerging Scholars****ix. List of Tables (if Any)**

This constitutes all the tables produced during the study. They are serially and consecutively numbered and are listed according to their numbers, title, and the specific pages where they may be located in the report. An example of a list of tables is shown in Figure 7.

*Figure 7.***LIST OF TABLES**

|           |                                                                                                                                                                               |    |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| TABLE 1.  | Sample Distribution of secondary school teachers selected from ten selected public schools in Obubra Local Government Area.                                                   | 51 |
| TABLE 2.  | Mean, Standard deviation and variance of the independent and dependent variables of the study                                                                                 | 59 |
| TABLE 3.  | Summary of Population t-test result of secondary school teachers' job effectiveness level in Obubra Local Government Area.                                                    | 62 |
| TABLE 4.  | Summary of Correlation analysis of the relationship between arbitration strategy and secondary school teachers' job effectiveness.                                            | 64 |
| TABLE 5.  | Summary of Correlation analysis of the relationship between dialogue strategy and secondary school teachers' job effectiveness.                                               | 66 |
| TABLE 6.  | Summary of Correlation analysis of the relationship between effective communication strategy and secondary school teachers' job effectiveness.                                | 68 |
| TABLE 7.  | Summary of Correlation analysis of the relationship between smoothing strategy and secondary school teachers' job effectiveness.                                              | 70 |
| TABLE 8.  | Summary of Multiple Regression results of the influence of arbitration, dialogue, effective communication, and smoothing strategies on secondary teachers' job effectiveness. | 72 |
| TABLE 9.  | Analysis of Variance (ANOVA) of the Regression Analysis                                                                                                                       | 74 |
| TABLE 10. | Relative Contributions of the four strategies to secondary school teachers' job effectiveness                                                                                 | 76 |

**x. List of Figures (if Any)**

Where figures like graphs, maps, diagrams, photographs or plates have been produced during the study and reported in any work, they have to be listed, showing their titles and the pages where they can be located in the report.

## **xi. List of Abbreviations (if Any)**

This is a list where abbreviations used in the study are listed and explained for clarity.

## **CHAPTER ONE: INTRODUCTION**

Chapter one of the report deals with the “introduction” to the study. It is the gateway to the whole study (Isangedighi & Bassey, 2012). Chapter one is “an introduction to the whole research, which is reported from chapters two to five. Chapter one is a window to the whole work” (Asim et al., 2017, p.187). In quantitative research, chapter one usually gives information under the following subsections:

### **i. Background to the Study**

The essence of this subsection is to provide the necessary background information that will show the readers the conditions, circumstances and factors that have given rise to the problem under investigation. Here, the researcher presents how deep-seated the problem under investigation is and, if not adequately addressed, may lead to more unpleasant consequences. It is also in the background of the study that both dependent and independent variables are explained and linked up correctly.

### **ii. Theoretical Framework**

A theoretical framework is a group of related ideas that guides a research project (Asim et al., 2017). It establishes the perspective through which the researcher views the problem. It provides the rationale for research to investigate a particular problem (Mertler & Charles, 2008). It provides the theoretical base that supports the researcher’s investigation and offers the reader a justification for the study. Here, the researcher is expected to identify a theory (or theories), mention who propounded it, the year it was propounded, state the tenets of the theory, and link the variables in the study with the tenets of the theory. The identified theory or theories must have relevance to the study’s variables.

### **iii. Statement of the Problem**

This is where the researcher should clearly express his intention of the study so well as to leave no reader in doubt about what the work is trying to address. In other words, the focus here is to present the actual problem to which one is trying to contribute a solution in clear terms and present the problem by elaborating,

**Basics of Research Report Writing for Behavioural Science Students and Emerging Scholars**

clarifying and delineating its scope. The statement of the problem can be stated in either a declarative or an interrogative form. In whichever form the problem is stated, it should be unambiguous and short (not exceeding a page of an A-4 paper).

**iv. Purposes of the Study**

Here, what the researcher intends to accomplish in the study is clearly stated or outlined. The purpose of the study should be stated in general and specific terms. In most cases, each purpose of the study is linked directly with a research question or hypothesis, where applicable.

**v. Research Questions**

Research questions refer to the major questions the researcher seeks to answer in the investigation. Good research questions would arouse and create sustained interest in the researcher to seek answers to such questions. Note that good research questions should be empirically answerable, not lend themselves to a yes or no answer, and be framed to guide the study's conduct. An example of a research question is *how much do males differ from females in their mathematics achievement mean scores?* Another example is *to what extent are school infrastructural facilities available in tertiary institutions?* Other examples of research questions can be found in Figure 8.

*Figure 8.*

- i. What is the level of secondary school teachers' job effectiveness in Obubra Local Government Area?
- ii. How does arbitration strategy relate to secondary school teachers' job effectiveness?
- iii. How does dialogue strategy relate to secondary school teachers' job effectiveness?
- iv. How does effective communication strategy relate to secondary school teachers' job effectiveness?
- v. How does smoothing strategy relate to secondary school teachers' job effectiveness?
- vi. How does arbitration, dialogue, effective communication, and smoothing strategies jointly influence secondary school teachers' job effectiveness?

## **vi. Statement of Hypotheses**

Hypotheses refer to intelligent guesses or hunches that the researcher formulates to guide his research for a solution to the problem. Hypotheses reflect measurable variables. They should be stated in a measurable and testable form. They should be stated in declarative form and tested using an inferential statistical test. It is essential to point out that some types of research, for instance, a descriptive survey, may require research questions with no compulsion for testing hypotheses. Hypotheses may be stated in alternative form or null form. An example of an alternative hypothesis is that *there is a significant difference in the mean scores of males and females in mathematics*. In null form, the same hypothesis may be stated as *there is no significant difference in the mean scores of males and females in mathematics*.

## **vii. Significance of the Study**

This subsection provides the justification or rationale for conducting the research. It tries to establish whether or not the study is of some value. It argues why your research matters, how it impacted your field, what it added to the body of knowledge, and how it will benefit others. The researcher should also explain the study's usefulness in advancing already-known theories, positions and results. The practical significance of the study should also be discussed. This implies that the researcher must identify all the stakeholders likely to benefit from the research outcome. In other words, the researcher should emphasise practical benefits derived from the study's findings by different stakeholders, one after the other. Thus, your research's significance, value, and contribution should be made clear to the reader in your academic writing.

## **viii. Assumptions of the Study**

Assumptions in research are out of the researcher's control, but if they disappear, your findings will be invalid, or your study will become irrelevant (Simon, 2011). For example, if you are conducting an empirical study, you need to assume that the variables involved in the study are measurable. If you are choosing a sample, you need to assume that this sample is representative of the population you wish to make an inference. It has been submitted by Leedy and Ormrod (2010) that "assumptions are so basic that, without them, the research problem itself could not exist" (p.62). Every research (behavioural science research) is carried out with some basic assumptions that border principally on the variables and sample of the study. For the variable, the basic assumptions are that the variable is measurable and normally distributed. For the sample, the basic assumptions are that the sample selection must be free of bias (fair) and the sample should be representative of the population of the study.

## **ix. Scope of the Study (or Delimitation)**

The scope/delimitation of the study refers to those characteristics that limit the coverage and define the boundaries of your study. The delimitations are in your control. The scope factors include the problem of the study, the choice of the objectives, the research questions/ hypotheses, variables of interest, theoretical perspectives to adopt and the population you choose to investigate. Here the researcher needs to set a boundary for his study correctly. Things to consider are the variables and substances of interest, the sample and the study area. When the scope of the study is appropriately delineated, it makes the discussion of findings and generalisation realistic and meaningful.

## **x. Limitations of the Study**

Limitations of a study are potential weaknesses in your study. We find limitations in almost everything we do. Suppose you are using a sample of conveniences instead of a random sample; then, the results of your study cannot be generally applied to a larger population, only suggested. If you look at one aspect, achievement tests, the information is only as good as the test itself. Alternatively, if you use a questionnaire to collect information when the most appropriate data collection strategy would have been using an observational schedule, that is a limitation. Another limitation is time. A study conducted over a specific time interval is a snapshot dependent on conditions occurring during that time (Simon, 2011). In this subsection, you must explain how you intend to deal with the limitations you are aware of so as not to affect the study's outcome. Limitations of the study usually focus on weaknesses of design, method and procedure that will limit the validity and generalisation of findings.

## **CHAPTER TWO: LITERATURE REVIEW**

This chapter reviews all literature about the research theme, usually organised under relevant subheadings based on research questions or hypotheses. No matter how innovative or novel a research study may appear, earlier researchers may have done something related to that research or in certain aspects of the research to which the new study attempts to contribute. A literature review relating to the field of interest is necessary to avoid any attempt to reinvent the wheel and waste precious resources (Asim et al., 2017). Here, the researcher is expected to do the following, among others (Ugodulunwa, 2016):

***Basics of Research Report Writing for Behavioural Science Students and Emerging Scholars***

- Review variables in the research topic and problem as well as relevant theories and empirical studies
- Organise and present literature reviewed coherently by ensuring each sentence is linked to the following and paragraphs are linked to one another using link words or phrases.
- Critique the studies regarding their weaknesses in design, method and procedure.
- Comment on the implications of the literature reviewed for your study.
- Cite current literature in the review; and
- End with a summary of the literature reviewed and clearly show the gaps to be filled.

**CHAPTER THREE: RESEARCH METHODOLOGY**

This chapter deals with the investigation procedure and consists of different sub-sections, depending on the type of investigation (research). Generally, it describes the design, sampling techniques, data collection techniques, and data analysis strategies as follows:

**i. Research Design**

Here, the researcher is expected to relate the design to the general approach adopted in the study. The researcher should mention the design, describe it, and justify why and how it is adopted in the study. The research design chosen and adopted should be appropriate and have high internal or external validity.

**ii. Area of the Study**

Here, the researcher is expected to locate the study area in terms of latitudes and longitudes (geographically and sometimes with demographic descriptions). This could be stated in the country, state, education, political/administrative zone, local government area etc., as the case may be. The researcher is also expected to indicate the study area's geographical borders (regarding neighbours).

**iii. Population of the Study**

Here, the researcher should mention the population of the study and describe its characteristics and context. A population may be finite if all items or subjects that constitute it can be counted. It is infinite if the size (number) cannot be counted.



***Basics of Research Report Writing for Behavioural Science Students and Emerging Scholars***

The population should be described explicitly in terms of geographical limits, demographic and sociological characteristics, the period covered, etc., as dictated by the problem of the study. The description should be such that it would leave no doubt in the readers' minds as to who is qualified or not qualified to be a member of the population under study. If possible, state the total members of the population.

**iv. Sampling Techniques**

Mention the sampling technique(s) to be adopted and describe it clearly. Describe how the sampling technique is applied in sample selection. Justify the choice of the sampling techniques and present the distribution of the population of the study in a table. Before drawing the sample, the researcher must define his unit of analysis or study. Who is being studied, and what constitutes the population from which the sample will be drawn? Other considerations that must be made when deciding on the sampling technique are the nature of the population, its characteristics, sub-groups and size. The researcher must also decide whether probability or non-probability sampling methods will be more suitable and justify the choice using literature support. The researcher must determine whether it is feasible to implement specific sampling strategies in selecting respondents based on factors such as the geographical spread of the population, access to the sample, available time, financial cost, etc. The type of research method pursued/adopted for the study is another determinant. Furthermore, the need to ensure unbiasedness in selecting participants must also be considered. For example, a researcher may use a proportionate stratified random technique for his study. The researcher may choose this method because he wants to give every member of the population an equal opportunity to participate (randomisation). Secondly, his population of interest contains males and females; thus, he stratifies it along gender lines. Thirdly, he wants to select his sample based on the proportion of individuals in both the male and female categories. Doing this will not only yield an unbiased sample but will promote fairness.

**v. Sample**

The sample (that portion of the population from which data were collected) is defined here as how it can be obtained. It may or may not be a representative part of the population. The proportion or percentage of the population selected as the study sample should be clearly stated, and the sample distribution should be presented in a table. Justify your decision to choose a particular sample size given your study's anticipated or desirable precision. When deciding on the sample size for a study, special attention must be given to the statistical power to be achieved/required; the allowance for possible attrition, missing data or non-response, the

***Basics of Research Report Writing for Behavioural Science Students and Emerging Scholars***

number of variables in the study; the type of research pursued, the number of sub-groups in the population to avoid sample over- or under-representation; the method of statistical analysis to be used; the size of the population; the expected response rate; the research timeframe; the funds at the researchers' disposal; the expected impact or utility value of the study. For example, a researcher might decide to study an entire population if it is small or manageable. In another situation, a researcher may require a large sample size due to statistical analytic requirements such as structural equation modelling (SEM). In contrast, another researcher may recruit a small sample for his study because it is qualitative.

**vi. Instrumentation**

This involves a description of the instrument(s) used in collecting the data. Such a description should include how the instrument was developed and the major features of the instrument. If the instrument used is constructed by you, describe clearly, with accurate details, how the set of items used to elicit responses for each variable was constructed. If the items were adapted from an existing instrument, summarise the procedure used in its construction and measurement properties. If the instrument was adopted, say so. In behavioural sciences, different instruments can be used for data collection, such as questionnaires, interviews, observation, and tests.

**vii. Validity and Reliability**

The validity of the instrument(s) used for the study should be established. Mention the type of validity established (whether face, content, predictive or construct) and justify the validity choice. Describe clearly the procedure you adopted for establishing the validity, and present the results of validity analysis (where necessary). Ensure that the instrument measure what it is supposed to measure.

An instrument is reliable if it consistently gives the same or similar results. This also should be established for the instrument (s). Mention the type of reliability estimate (whether internal consistency, stability or equivalence) and justify your choice. Describe the procedure for establishing the instrument's reliability (split-half, test-retest, Cronbach Alpha, etc.). Mention the criteria or standards from the literature for judging the reliability level of the instrument.

**viii. Procedure for Data Collection**

Here, the researcher reports the steps to collect pertinent data for the study. This entails whether the researcher collected the data, with the help of field assistants, by mail, etc. The researcher should state how authorities sought permission to collect

***Basics of Research Report Writing for Behavioural Science Students and Emerging Scholars***

data from the population as the sample. Describe how research assistants (if any) were trained and used for data collection. Describe clearly how each instrument was administered. Mention the duration of data collection. Describe how data from each instrument were scored. In the case of an experimental study, the researcher should also describe pre-test administration, treatment package/ intervention/ programme, treatment given to the experimental group, package given to the control group and how sources of internal invalidity were controlled. Treatment should last for at least eight weeks for sufficient impact to be made (Ugodulunwa, 2016).

**ix. Procedure for Data Preparation and Scoring**

In research that generates an extensive mass of data, the researcher needs to state beforehand how he plans to process the data collected/ generated. Data processing involves transferring collected data into coded data for further processing through a data processing instrument, reliably through computer statistical software for data management and analysis (Owan & Bassey, 2018). The researcher also describes how he scored the participants' responses variable by variable. States how he coded or transformed the data in preparation for analyses.

**x. Procedure for Data Analysis**

The procedures for analysing collected data deal with the technical procedures needed for statistical or quantitative analysis. The student/ researcher should reflect on the statistical techniques appropriate for the research design. The most appropriate mode of statistics appropriate for the data collected must be used. Note that the choice of each technique depends on the nature of the data (nominal, ordinal, interval or ratio). Present data analysis techniques you used for each research question and/or hypothesis; describe how and why each technique is applied. If the data are not many, simple frequency count, for example, can be used, and statistical computation is done manually. The data analysis techniques may be the elementary application of measures of central tendency, variability, regression, correlation, time series, ANOVA, contingency analysis or more complicated techniques involving programming etc. Results from the data analysis should be presented appropriately in charts, graphs, etc., describe the method(s)/mode (s) used, or cite sources that do. Report the statistical analysis, including data sources and the name of any computer software packages used.

## **xi. Operational Definition of Research Variables**

Here, the student or researcher should define only terms whose meanings or methods of obtaining measurement are subject to ambiguity in measurable terms. Critical variables in the research topic, research questions and hypotheses need to be operationally defined in how they are measured in the study. Definitions of terms should be either conceptual or operational.

## **xii. Ethical Considerations**

Ethical concerns refer to the regulations that outline the appropriate and inappropriate actions for researchers to take when conducting their studies. While conducting your study, understanding the governing ethical principles is important to guide the research conduct. Therefore, researchers must indicate how relevant ethical principles were considered and followed in the research. In their text, Cohen et al. (2018) listed the following as some ethical issues that must be considered in behavioural research: informed consent; confidentiality and anonymity; identification and non-traceability; non-maleficence; beneficence and duty of care; responsibilities (for what and to whom); gaining access; overt and covert research; disclosure and public versus private knowledge; relationships and differential power relations; interest at stake in the interest; rights, permissions and protections; ownership and control of data; access to data and its archiving; the role and powers of research sponsors.

## **CHAPTER FOUR: RESULTS AND DISCUSSION**

### **i. Results**

This chapter embodies the results or findings and discussion of the study. Here, you present your research results in tables and/or graphs according to the research questions and/or hypotheses formulated for the study. Note that only the summaries of such analyses in the form of tables, figures and charts are presented in this chapter. Each table, figure or chart should have a title and number. The number should be in the Arabic numeral and not Roman. For tables, the number and title should appear on top of the table, while for figures, the number and title should appear below the figures. It is advisable to present the results according to the research questions or hypotheses to which they relate. The results so presented are duly interpreted. After statistical interpretations have been given to data analysis, there is a need to explain clearly what the statistical result means.

***Basics of Research Report Writing for Behavioural Science Students and Emerging Scholars*****ii. Discussion of Findings**

How the outcome/findings of the present study fit into previous research, as highlighted in the areas of literature review, becomes evident at this point. Here, the researcher tries to advance possible interpretations and explanations of his findings and relates these to previous research findings. Implications of the findings, that is, how the present findings relate to specific uses, theories, or practise, can also be discussed here. The limitations of generalising study findings should also be discussed, with good suggestions for future researchers.

**CHAPTER FIVE: SUMMARY, CONCLUSION AND RECOMMENDATION**

This chapter is a recapitulation of what the research set out to do and the outcome of the research undertaking. It has the following subsections:

**i. Summary of the Study**

This subsection has to do with a restatement of the purpose of the study, a brief explanation of the methodology adopted and an outline of significant findings. Summary of the study provides a snapshot of the entire research process, including the study's objectives, theories used, the hypotheses formulated, the methods used (including instruments, sampling methods, data collection methods etc.), and an overview of the results/findings reached.

**ii. Conclusion**

The conclusion is the inference that may be drawn from the study based on the major research findings. In conclusion, the research has to give some impact implications of the study's outcome regarding the original problem that necessitates the research. The research should be able to answer a question like: "giving these findings, what next?" (Isangedihi & Bassey, 2012). Thus, the conclusion does not restate the findings already highlighted in chapter four or elsewhere in this report. In other words, a conclusion is drawn based on the study's findings and is usually stated assertively. A reasonable conclusion should also clearly state the implications of the study's findings for policy and practical reforms.

***Basics of Research Report Writing for Behavioural Science Students and Emerging Scholars*****iii. Recommendations**

Suggested actions could be taken in light of findings to bring about improvements in the system or the discipline. The recommendations should be based on major research findings. Sometimes it may be necessary to direct such recommendations to different groups of beneficiaries or stakeholders of the research results, for example, policymakers, teachers, students, parents and society at large, depending on the research outcome.

**iv. Suggestions for Further Research**

Since research is a continuous process, it follows that the end of one study marks the beginning of another. To ensure this, it is necessary to make suggestions at the end of each study on areas on that future research could be based. Such suggestions might take care of the new areas/ideas that emerged during the research or the literature review. Suggestions could also be made for replication of the study, change of population or sample, the addition of study variables, etc.

**ENDING PAGES****i. References**

The references section lists all sources cited in the work, such as textbooks, journals, conference/seminar papers, theses, indexes, mimeographs, etc., used during the study. Therefore, all works cited in the body of the report should be listed in the references to enabling other researchers to identify and locate such sources. For references, it is important to consider very recent sources (works within the past five years) for empirical studies. However, for theories and methodology, seminal works are highly recommended; thus, old literature may be used for these sections. Only cite works that are very useful and related to the study at hand to provide a basis for your work. Before citing any source in your writing, ensure it has complete bibliographic information. Therefore, use such information to prepare proper referencing of the material before citing to avoid citation issues without corresponding references. The latest referencing style should be used for citation in-text and compilation of reference lists as per the requirement of the particular journal.

**Basics of Research Report Writing for Behavioural Science Students and Emerging Scholars****ii. Appendices**

This subsection consists of supplementary materials, which in the researcher's opinion, need not be included in the body of the report but could be crucial to the reader. An appendix contains samples of an instrument(s) used for the study and detailed calculations made to arrive at the final results. The materials could be research instruments, correspondences made during the study, list(s) of units studied, etc. Appendix headings are in boldface, centred, at the top margin. Allow two block lines between the heading and first entries (Ndiyo, 2005). Appendices are labelled A, B, C or I, II, III etc. Different types of material should be in a separate annexe, and each type of material should be complete in and of itself. The appendices follow the reference list.

**CONCLUSION**

In research report writing, students or researchers make serious mistakes that could cause total rejection of their thesis during the proposal defence or even during the final oral defence. According to Kinash (n.d), these mistakes include frequent use of long quotations, too many typographical and spelling mistakes, grammatical errors, statements, and generalisations not supported by literature evidence, plagiarism, and improper citation in text and references. Other errors could be using secondary sources, writing off-topics, exceeding the specific word count or pages in some sections of the thesis, fabricating, falsifying and misrepresenting data, and non-adherence to acceptable style/format and thesis guidelines, generally. Students should, therefore, guard against these pitfalls in writing their research projects, theses or dissertations for graduation.

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**ADDITIONAL READING**

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**KEY TERMS AND DEFINITIONS**

**Citation:** Citation simply refers to an acknowledgement given to an author or a source for using their work. The aim of citation is to let readers know the original source where a piece of information presented in a document has been derived.

**Data:** Data are raw facts that are gathered from the research awaiting processing for relevant meaning associated with them to be uncovered. When data are processed, meanings can be made from them for a thorough understanding of the event, phenomena or trait about which the data was collected.

**Emerging Scholars:** Emerging scholars are individuals undergoing training at a tertiary institution that are venturing or intending to venture into research, writing and publication. These can include undergraduates, postgraduates or doctoral students and graduates that have yet to become major authorities in their fields.

**Research Methods:** This is a generic term used to refers to a compendium of all the procedures, tools, approaches and materials used in pursuing a given research

***Basics of Research Report Writing for Behavioural Science Students and Emerging Scholars***

project. Basically, research methods can be classified into mono-methods and mixed methods.

**Research Reporting:** This is the last stage in the research process where researchers communicate the results of their studies to the wider community. Research reports can be prepared as journal articles, chapters in books, theses/dissertations, books, etc.

**Research:** Research is an action-oriented exercise and a systematic means of acquiring knowledge through the collection, preparation, analysis and interpretation of data to reach meaningful conclusions.

**Sample:** This is a portion of the population selected (using one or more acceptable strategy) for an investigation. A sample can be homogenous or heterogenous; it can be small or large; it can be representative or non-representative. However, the information derived from a sample about a phenomenon of interest can be generalised to the population.

**Sampling:** This is the process of selecting a sample for a research project. It can also be a collection of procedure followed in recruiting a sample for a study. Sampling procedures can be probability-based or non-probability-based, with each having their strengths, weaknesses and suitable situations.